<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td rowspan="11">1</td><td>Defining the problem</td><td></td></tr>
<tr><td>Vary p and measure B OR p is the independent variable and B is the dependent variable.</td><td>1</td></tr>
<tr><td>Keep V constant or potential difference between the ends of each conductor constant.</td><td>1</td></tr>
<tr><td>Methods of data collection</td><td></td></tr>
<tr><td>Labelled diagram of workable experiment including:
• conductors in parallel connected in series to power supply and resistor
• circuit symbols for (variable) resistor and power supply
• X labelled and one other label from L, P, Q, p and q.</td><td>1</td></tr>
<tr><td>Voltmeter connected in parallel with conductors (to measure V) and conductors in parallel connected to a power supply.</td><td>1</td></tr>
<tr><td>Method to measure L and p and q e.g. use a rule / ruler / calipers.</td><td>1</td></tr>
<tr><td>Method to measure B, e.g. use a (calibrated) Hall probe and adjust / rotate probe until maximum value.</td><td>1</td></tr>
<tr><td>Method of Analysis</td><td></td></tr>
<tr><td>Plots a graph of B against $\frac{1}{p}$ or equivalent.
Do not accept logarithms.</td><td>1</td></tr>
<tr><td><table><tr><td>B against $\frac{1}{p}$</td><td>$\frac{1}{p}$ against B</td></tr><tr><td>$Y = \frac{L \times \text{gradient}}{AV}$</td><td>$Y = \frac{L}{AV \times \text{gradient}}$</td></tr></table></td><td>1</td></tr>
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Question	Answer	Marks
1	<table><tr><td>B against $\frac{1}{p}$</td><td>$\frac{1}{p}$ against B</td></tr><tr><td>$Z = \frac{q \times y\text{-intercept}}{\text{gradient}}$ OR $Z = \frac{Lq \times y\text{-intercept}}{YAV}$</td><td>$Z = -q \times y$-intercept</td></tr></table>	**1**
	Additional detail including safety considerations Any **six** from:	**6**
	D1 precaution linked to <u>high current</u> / <u>hot conductors</u>, e.g. use gloves / switch off power supply when not measuring B / between measurements / allow conductors to cool	
	D2 keep A **and** L **and** q constant	
	D3 use calipers / micrometer to measure diameter / d of conductor **and** $A = \frac{\pi d^2}{4}$	
	D4 repeat measurements of d <u>in different positions</u> **and** <u>average</u> d	
	D5 method to determine the position of X in relation to the conductors, e.g. divide L by two to find the midpoint of P / Q **and** use a set square / protractor / plumb line to mark X OR divide L by two to find the midpoint of P / Q **and** use a grid to mark X	
	D6 measure B (using Hall probe) first in one direction and then in the opposite direction and average B OR Measure B with current / p.d. in one direction and then in the opposite direction and average B	
	D7 additional detail on measuring p and / or q, e.g. measure to the conductor and add on the radius	

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Question	Answer	Marks
1	D8 description of method to keep *q* constant, e.g. tape / adhesive putty to fix conductor Q to the bench OR for vertical methods fix conductor Q in clamp(s) attached to stand(s) to keep *q* constant	
	D9 method to keep P and Q parallel, e.g. measure the separation (between the conductors) at different points	
	D10 relationship valid <u>if</u> a straight line is produced (with a *y*-intercept = $\frac{YZAV}{Lq}$). Do not accept passing through the origin.	
	D11 method to keep *V* constant, e.g. adjust / change variable resistor / power supply to keep voltmeter reading constant.	

Question	Answer	Marks
2(a)	gradient = $-\frac{1}{K}$ *y*-intercept = $\ln(\theta_0 - \theta_R)$	**1**